Cell – The Building Block of Life

A cell is the basic **structural and functional unit** of life.

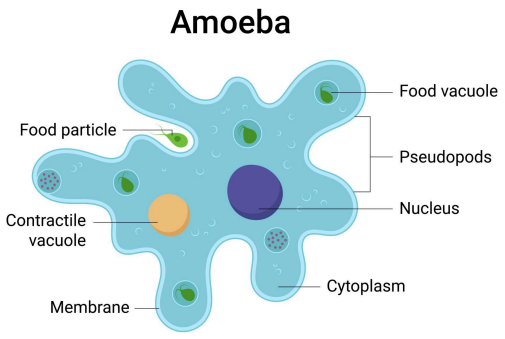
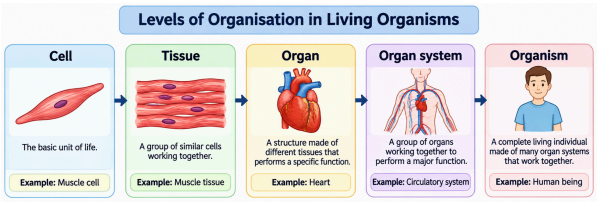
- **Structural unit**: All living organisms are made up of one or more cells.
- **Functional unit**: Cells carry out essential life processes such as respiration, nutrition, growth and excretion.
- A cell is the smallest unit of life capable of **performing essential life functions**.

Examples: Bacterial cells, plant cells and animal cells.

Where do we find cells?

Cells are present in all **living organisms, including plants, animals, fungi and microorganisms.**

Some organisms consist of only **one cell,** while others consist of **many cells**.

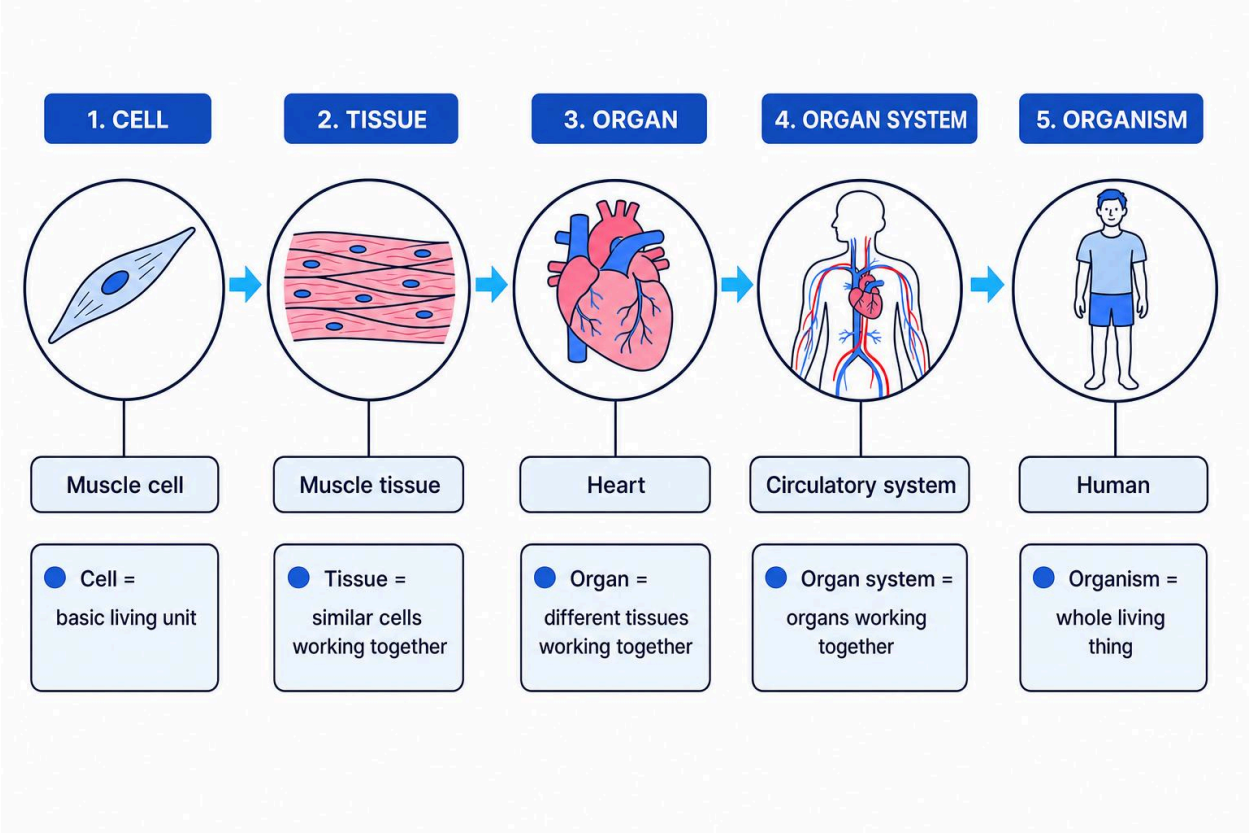
Unicellular organisms	Multicellular organisms
Made up of one cell	Made up of many cells
One cell performs all essential life functions	Different cells may perform specialised functions
Examples: Amoeba, bacteria	Examples: Humans, trees, dogs
	

Cells perform different life processes necessary for survival.

- **Nutrition**: Obtain or produce nutrients.
- **Respiration**: Release usable energy from food.
- **Growth**: Increase in size and produce new cell material.
- **Excretion**: Remove metabolic waste.
- **Reproduction**: Many cells divide to form new cells.
- **Response**: Detect and respond to changes in their surroundings.

In multicellular organisms, specialised cells work together to perform different functions.

Levels of organisation in living organisms



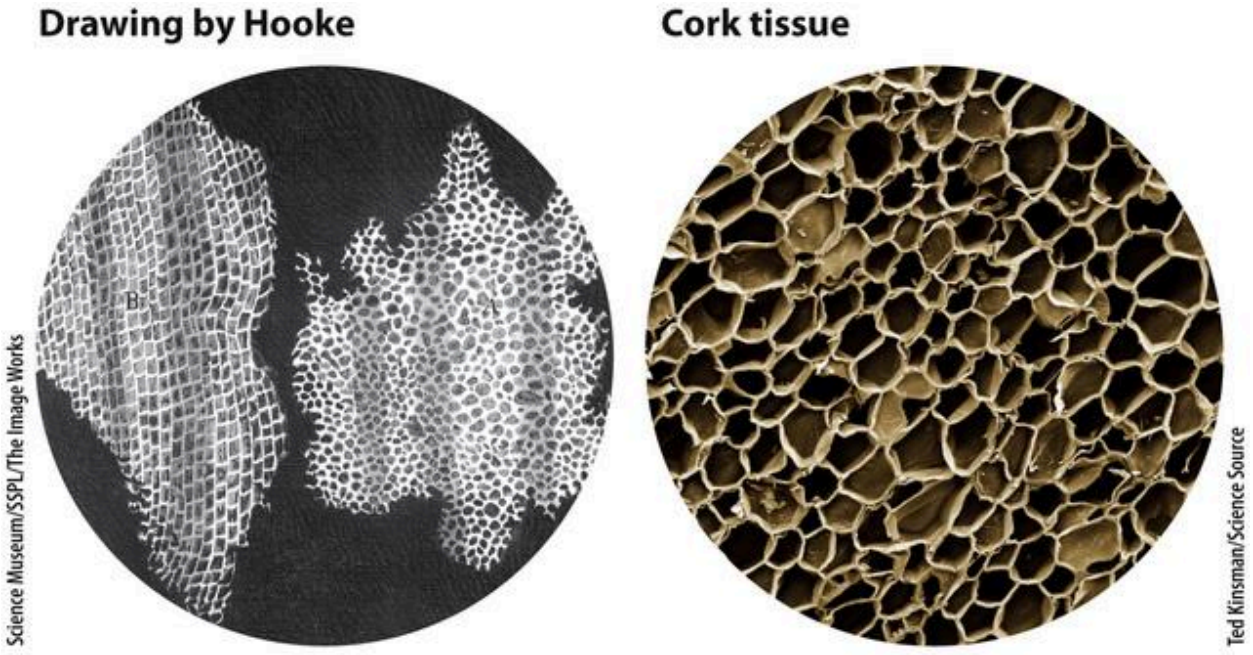
Definitions:

- **Cell:** Basic structural and functional unit of life.
- **Tissue:** A group of similar cells working together to perform a function.
- **Organ**: A structure made of different tissues working together.
- **Organ system:** A group of organs working together to perform a major function.
- **Organism**: A complete living individual.

Discovery of the Cell

Robert Hooke – 1665

- Robert Hooke observed a thin slice of cork using a microscope he had designed.
- He noticed many small, box-like compartments.
- He named these compartments cells.
- The structures he observed were the walls of dead plant cells.



Scientists and Their Contributions

Scientist	Year	Contribution
Robert Hooke	1665	Observed cork cells and named them cells.
Antonie van Leeuwenhoek	1674	Observed living microorganisms in pond water.
Robert Brown	1831	Described the nucleus in plant cells.
J. E. Purkinje	1839	Introduced the term protoplasm.

Cell Theory

Cell theory explains the relationship between cells and living organisms.

The three main principles of cell theory are:

1. All living organisms are made up of one or more cells.
2. The cell is the basic structural and functional unit of life.
3. All cells arise from pre-existing cells.

Scientists who contributed to cell theory

Scientist	Contribution

Matthias Schleiden	Proposed that all plants are made up of cells.
Theodor Schwann	Proposed that all animals are made up of cells.
Rudolf Virchow	Popularised the principle that new cells arise from existing cells.